

```

component fortress.executable
export Executable
* A comment-to-end-of-line
run():() = do
  needleLength = 20
  numRows = 10
  tableHeight = needleLength numRows
  var hits:ℝ64 = 0.0
  var n:ℝ64 = 0.0

  println("Starting parallel Buffons")
  recordTime(6.0)
  for i ← 1 # 3000 do
    δX = random(2.0) - 1
    δY = random(2.0) - 1
    rsq = δX2 + δY2
    if 0 < rsq < 1 then
      y1 = tableHeight random(1.0)
      y2 = y1 + needleLength (δY/sqrt(rsq))
      (yL, yH) = (y1 MIN y2, y1 MAX y2)
      if ceiling  $\left(\frac{y_L}{\text{needleLength}}\right)$  = floor  $\left(\frac{y_H}{\text{needleLength}}\right)$  then
        atomic do hits += 1.0 end
      end
      atomic do n += 1.0 end
    end
  end
  probability =  $\frac{\text{hits}}{n}$ 
  πest =  $\frac{2.0}{\text{probability}}$ 
  printTime(6.0)
  println("")
  print("estimated Pi = ")
  println(πest)
end
end

```